

NanoGT Match Report: Maple syrup urine disease

Disease: Maple syrup urine disease (ORPHA:511)

Primary gene: BCKDHA

Gene CDS: 1335 bp

Inheritance: Autosomal recessive

Target tissues: liver, CNS

Top 5 GT Precedent Matches

Rank	Program	Vector	Score	Confidence	Approval
1	OAV101-IT	AAV9	8.0/10	High	approved
2	Zolgensma	AAV9	8.0/10	High	approved
3	Luxturna	AAV2	7.8/10	High	approved
4	DTX301	AAV8	7.3/10	Medium	phase2
5	BMN 307	AAV5	7.1/10	Medium	phase2

Match #1: OAV101-IT

Precedent disease: Spinal Muscular Atrophy

Vector: AAV9

Tissue target: CNS/spinal cord

Composite score: 8.0 / 10

Score Breakdown

Dimension	Score	Max
Packaging fit	2.0	2.0
Tissue tropism	2.0	2.0
Protein class	1.5	2.0
Pathway similarity	0.5	2.0
Inheritance compatibility	1.0	1.0
Approval precedent	1.0	1.0

Rationale

- Gene CDS 1335bp / cargo 4700bp (28% utilized)
- Vector tropism plus precedent target match: cns
- Both intracellular proteins
- Inheritance match (Autosomal recessive <-> AR)
- Different pathway (mitochondrial_complex vs motor_neuron)
- Approval status: approved

Match #2: Zolgensma

Precedent disease: Spinal Muscular Atrophy

Vector: AAV9

Tissue target: CNS/motor neuron

Composite score: 8.0 / 10

Score Breakdown

Dimension	Score	Max
Packaging fit	2.0	2.0
Tissue tropism	2.0	2.0
Protein class	1.5	2.0
Pathway similarity	0.5	2.0
Inheritance compatibility	1.0	1.0
Approval precedent	1.0	1.0

Rationale

- Gene CDS 1335bp / cargo 4700bp (28% utilized)
- Vector tropism plus precedent target match: cns
- Both intracellular proteins
- Inheritance match (Autosomal recessive <-> AR)
- Different pathway (mitochondrial_complex vs motor_neuron)
- Approval status: approved

Match #3: Luxturna

Precedent disease: Leber congenital amaurosis type 2

Vector: AAV2

Tissue target: retina/RPE

Composite score: 7.8 / 10

Score Breakdown

Dimension	Score	Max
Packaging fit	2.0	2.0
Tissue tropism	0.3	2.0
Protein class	1.5	2.0
Pathway similarity	2.0	2.0
Inheritance compatibility	1.0	1.0
Approval precedent	1.0	1.0

Rationale

- Gene CDS 1335bp / cargo 4700bp (28% utilized)
- No tissue overlap (disease: ['liver', 'CNS'], vector: [])
- Both intracellular proteins
- Inheritance match (Autosomal recessive <-> AR)

- Pathway match: retinal_visual_cycle
- Approval status: approved

Match #4: DTX301

Precedent disease: Ornithine transcarbamylase deficiency

Vector: AAV8

Tissue target: liver

Composite score: 7.3 / 10

Score Breakdown

Dimension	Score	Max
Packaging fit	2.0	2.0
Tissue tropism	2.0	2.0
Protein class	1.5	2.0
Pathway similarity	0.5	2.0
Inheritance compatibility	0.7	1.0
Approval precedent	0.6	1.0

Rationale

- Gene CDS 1335bp / cargo 4700bp (28% utilized)
- Vector tropism plus precedent target match: liver
- Both intracellular proteins
- LOF inheritance — compatible for gene replacement
- Different pathway (mitochondrial_complex vs urea_cycle)
- Approval status: phase2

Match #5: BMN 307

Precedent disease: Phenylketonuria

Vector: AAV5

Tissue target: liver

Composite score: 7.1 / 10

Score Breakdown

Dimension	Score	Max
Packaging fit	2.0	2.0
Tissue tropism	1.5	2.0
Protein class	1.5	2.0
Pathway similarity	0.5	2.0
Inheritance compatibility	1.0	1.0
Approval precedent	0.6	1.0

Rationale

- Gene CDS 1335bp / cargo 4700bp (28% utilized)

- Precedent target match: liver
- Both intracellular proteins
- Inheritance match (Autosomal recessive <-> AR)
- Different pathway (mitochondrial_complex vs amino_acid_metabolism)
- Approval status: phase2